

JOURNAL OF UNIVERSAL INVARIANTS

*Foundations of Recursive Compression,
Relational Structure, and Explanatory Horizons*

Volumes I – III | Universal Invariant Compiler Protocol

Editorial Introduction

This journal documents the progressive identification of structures that survive translation across domains. The research program operates under a single methodological constraint: no claim is retained unless it survives repeated compression across independent domains. Claims that reduce to existing mathematics are mapped to their prior art. Claims that do not are held as open problems, not conclusions.

Most scientific disciplines study local representations of phenomena native to their domain. This program studies what remains when those representations are removed. The residue, if any survives, is a candidate invariant.

The articles collected here are not independent contributions. They constitute a single recursive audit conducted across physics, biology, information theory, cybernetics, control theory, and adaptive systems. Each volume extends the compression one layer deeper than the last.

The objective is not a Theory of Everything. The objective is precise identification of the deepest structure currently reachable by the compiler. That structure is named at the end of Volume III. Whether anything deeper exists is listed as an open problem.

Readers familiar with viability theory (Aubin), relational biology (Rosen), and categorical systems theory will recognize structural overlap with those programs. Where overlap exists it is noted. The claim of this journal is not priority but synthesis.

VOLUME I

The Compiler

Recursive Compression, Kernel–Image Duality, and Adaptive Multiplicity

Volume I establishes the Universal Invariant Compiler Protocol as a formal research method and derives three foundational structures: the compression mapping, the kernel–image duality, and the adaptive multiplicity measure. These three structures recur without modification across every subsequent domain audit.

ARTICLE I

The Universal Invariant Compiler Protocol

Abstract

The Universal Invariant Compiler Protocol (UICP) is a recursive procedure for identifying structures that survive translation between domains. The protocol operates by iteratively removing implementation-specific vocabulary while testing whether operational functionality is preserved. Fixed points of this compression process are candidate invariants.

Formal Statement

Let a representation be any domain-specific description of a system. The UICP applies a compression mapping F to that representation, producing an observable image. The protocol asks: what survives?

Every representation admits a decomposition into five components:

$$\Sigma = (C, D, F, \ker(F), M^*)$$

Symbol	Meaning
C	Configuration space — all possible states of the system
D	Observable image — what is accessible to measurement
F	Translation operator — the compression mapping $C \rightarrow D$
$\ker(F)$	Kernel — configurations mapped to the same image
M^*	Adaptive multiplicity — dimension of the kernel

Principal Prediction

Systems possessing larger kernels exhibit greater capacity to reconstruct their functional output after perturbation. This prediction is domain-independent and is tested in every subsequent volume.

The decomposition Σ is not a new mathematical object. It is a named application of standard linear algebra to systems analysis. The contribution is the claim that this decomposition identifies the same invariant object across domains that have not previously been compared.

ARTICLE II

Kernel–Image Duality**Abstract**

Observations are not direct access to an underlying configuration space. Observations are images produced by compression operators. This article formalizes the consequences of that constraint.

Core Identity

By the first isomorphism theorem of abstract algebra, the quotient of a configuration space by the kernel of a mapping is isomorphic to the image of that mapping:

$$\Omega / \ker(F) \cong \text{Image}(F)$$

Interpretation

Two distinct configurations C_1 and C_2 may produce identical observable images D . This is not an epistemic failure. It is a structural property of any system governed by a many-to-one compression operator.

The consequence is precise: perfect reconstruction of the underlying configuration space from the observable image is generally impossible. Science therefore accesses $\Omega / \ker(F)$, not Ω itself.

OBSERVABILITY CONSTRAINT

Science accesses $\Omega / \ker(F)$, not Ω

This does not imply that the underlying domain is inaccessible in principle. It implies that access requires additional structure beyond what the compression operator alone provides. That additional structure is the subject of later volumes.

ARTICLE III

Adaptive Multiplicity and Degeneracy**Abstract**

Adaptive systems persist through perturbation because the functions they instantiate admit multiple realizations. This article formalizes that capacity as a measurable quantity.

Definition

$$M^* = \dim(\ker(F))$$

Adaptive multiplicity M^* measures the number of alternative configurations that produce the same functional output. A system with $M^* = 0$ has no alternative realizations. Perturbation that destroys its current configuration destroys its function. A system with $M^* > 0$ can reconstruct its function through an alternative configuration.

Domain Instances

Domain	M* Instantiation
Metabolic networks	Flux degeneracy — multiple enzyme pathways producing the same metabolic output
Neural circuits	Synaptic degeneracy — multiple connectivity patterns producing the same behavior
Ecological webs	Functional redundancy — multiple species performing the same trophic role
Software systems	Implementation equivalence — multiple code paths producing the same output
Quantum error correction	Code distance d — number of physical errors correctable before logical failure
Institutions	Procedural redundancy — multiple decision pathways reaching the same outcome

Core Principle

Persistence under perturbation increases when multiple relational configurations produce the same functional outcome. This is the domain-independent content of adaptive multiplicity.

M is not a new mathematical object. Kernel dimension is standard linear algebra. The contribution is the claim that kernel dimension is the correct cross-domain measure of what is informally called resilience, robustness, or adaptive capacity in domain-specific literatures.*

ARTICLE IV

Functional Invariance and Realization Equivalence

Abstract

Different physical implementations may realize identical functional structures. This article formalizes the conditions under which two distinct systems are functionally equivalent and states the consequence for system identity.

Principle

Let A and B be distinct configurations. They are functionally equivalent under operator T if:

$$A \neq B \quad \text{while} \quad T(A) = T(B)$$

Consequence for Identity

If functional equivalence under T is the relevant criterion, then the identity of a system is more strongly associated with the constraint-preserving transformation T than with its physical realization.

This is not a philosophical claim. It is a structural claim with measurable consequences: two systems that produce the same observable behavior under T are indistinguishable at the level of T, regardless of substrate differences.

Relation to Prior Art

This is the domain-independent content of Ho-Kalman realization theory in control engineering, which demonstrates that the minimal state-space realization of a system is unique up to similarity transformation. The claim here is that the same equivalence class structure appears in biological, physical, and institutional systems.

ARTICLE V

The Thermodynamic Ledger

Abstract

Every maintained distinction carries an energetic cost. This article formalizes that cost as a constraint on the persistence of any structured system.

Ledger Axiom

$$D \rightarrow \dot{W}$$

Any distinction D that is maintained against perturbation requires continual work $\dot{W}$. Distinctions are not thermodynamically free. A system that ceases to expend work on a distinction loses that distinction to entropic relaxation.

Consequence

Persistence requires energy expenditure. This is not a biological observation. It is a consequence of the second law of thermodynamics applied to any structured system maintaining a non-equilibrium state. The cost is proportional to the specificity of the distinction being maintained and the magnitude of the perturbative environment.

This article provides the thermodynamic grounding for the Cost of Optionality relationship noted in earlier work: maintaining a large kernel M^ under background perturbation requires proportionally greater energy expenditure. The two claims are consistent and the thermodynamic constraint bounds the achievable M^* for any physically realizable system.*

ARTICLE VI

Relation Before Substance

Abstract

Recursive application of the UICP to domain-specific ontologies consistently produces a residue in which relational structure survives compression longer than objects. This article documents that result and states its current status: a robust empirical finding of the compiler, not yet a derived theorem.

Compression Sequence

Applying UICP across physics, biology, and mathematics yields the following compression sequence:

$$\text{Particles} \rightarrow \text{Fields} \rightarrow \text{Geometry} \rightarrow \text{Information} \rightarrow \text{Distinction} \rightarrow \text{Relation}$$

At each stage, the preceding layer is expressible in terms of the succeeding layer, but not conversely. Objects reduce to fields, fields to geometric structures, geometric structures to information-theoretic constraints, information to distinguishability, and distinguishability to relational structure.

Hypothesis

Objects are intersections of relations. Relations are not compositions of objects. This reversal of the standard ontological order is consistent with the structure of category theory, in which morphisms are primitive and objects are derived, and with relational quantum mechanics, in which physical properties are defined relative to other systems rather than absolutely.

Status

This is a hypothesis supported by the compression result, not a proven theorem. The transition $? \rightarrow R$ remains unformalized and is listed in the Open Problems of Volume III.

ARTICLE VII

Distinction Formation

Abstract

Compression analysis reveals a boundary that the compiler cannot currently cross. All known theories presuppose distinctions. Few derive them from more primitive structure. This article names that boundary precisely.

The Boundary

$? \rightarrow D$

Every mathematical and physical framework encountered in the UICP audit takes distinguishability as given. Sets presuppose distinguishable elements. Probability theory presupposes distinguishable events. Quantum mechanics presupposes distinguishable outcomes. Category theory presupposes distinguishable objects and morphisms.

Open Question

What mathematical structure generates distinguishability itself without presupposing it? This is the deepest unresolved question identified by the compiler in Volume I. It is not answered here. It is stated precisely so that subsequent work can address it directly.

Symmetry breaking is the strongest existing candidate for a mechanism generating distinctions. Volume III investigates this candidate and finds that it partially explains distinction formation but does not eliminate the boundary, because defining symmetry requires specifying what may be transformed, which presupposes distinguishable states.

ARTICLES VIII – XII

Modeling, Self-Reference, and Explanatory Horizons

Summary

Articles VIII through XII trace the implications of distinction formation through a sequence of structural transitions. Each transition is stated as a formal claim with identified open problems. The sequence is:

$R \rightarrow D \rightarrow M \rightarrow S \rightarrow H$

Symbol	Structure and Derivation Status
R	Relational structure — deepest surviving invariant identified by compression; status: empirical finding
D	Distinction — emerges when equivalence classes become unstable; derivation from R: open problem
M	Model — arises when distinctions are organized into predictive structures; transition $D \rightarrow M$: partially formalized
S	Self-model — critical transition when a model contains a representation of itself; transition $M \rightarrow S$: formalized in recursive function theory
H	Horizon — self-inclusive systems encounter irreducible incompleteness limits; transition $S \rightarrow H$: formalized by Gödel (1931) and Turing (1936)

The Horizon Principle

Complete self-description is unavailable to any sufficiently expressive self-referential system. This is not a conjecture. It is a theorem with two independent proofs: Gödel's incompleteness theorems establish it for formal systems of sufficient expressive power, and Turing's halting problem establishes it for computational systems. The UICP result is that this horizon appears generically at the $S \rightarrow H$ transition in any system that develops self-modeling capacity.

Volume I — Summary Result

The collected audits of Volume I identify the following sequence as the strongest surviving invariant currently extractable by the UICP:

MAXIMAL INVARIANT — VOLUME I

$R \rightarrow D \rightarrow M \rightarrow S \rightarrow H$

This sequence represents the compression frontier at the close of Volume I. Each arrow is a structural transition. The derivation status of each transition is identified above. Two transitions are fully formalized ($M \rightarrow S$ in recursive function theory; $S \rightarrow H$ by Gödel and Turing). Three remain open problems ($? \rightarrow R$, $R \rightarrow D$, $D \rightarrow M$).

VOLUME II

The Spacetime Audit

Emergence of Geometry from Relational Structure

Volume I established the sequence $R \rightarrow D \rightarrow M \rightarrow S \rightarrow H$ as the current compression frontier. Volume II applies the UICP to spacetime itself, treating geometry not as a given substrate but as a candidate derived structure. The central question is whether spacetime appears as a primitive or as an emergent layer within the invariant sequence.

ARTICLE XIII

The Spacetime Hypothesis

Research Question

Is spacetime a fundamental substrate, or is it an emergent representation generated by deeper relational processes?

Competing Frameworks

Standard View	UICP Audit
Spacetime is the arena in which matter and observers exist	Spacetime is a derived object produced by relational systems
Observers are located within spacetime	Observers generate spacetime-like representations in order to model relational persistence
Geometry precedes matter	Relational structure precedes geometry

Prediction

If spacetime is emergent, then relational structure should be identifiable prior to geometric structure in the compression sequence. The audit below tests this prediction.

ARTICLE XIV

Relation Before Distance

Logical Priority Argument

Distance presupposes distinguishable objects at different locations. Distinguishable objects presuppose distinctions. Distinctions presuppose relational structure. Therefore distance cannot be ontologically prior to relation.

Distance → D → R

Reading the chain in the direction of logical dependence: distance requires distinction requires relation. Relation is therefore prior. This is a logical argument, not a physical one, and it holds independent of the specific physical theory of spacetime.

Consequence

Geometry is secondary. Relational structure is primary. This is consistent with the approach of relational quantum mechanics (Rovelli), causal set theory (Sorkin), and spin foam models, all of which derive geometric structure from combinatorial or relational primitives.

ARTICLE XV

Distinction Networks and Proto-Geometry

Research Question

What is the minimal structure capable of producing geometry from the relational primitives identified in Volume I?

Emergence Chain

R → D → Adjacency → Topology → Geometry

Distinctions between relational states generate adjacency structure: some states are more directly connected than others. Adjacency generates neighborhood structure. Neighborhood structure generates topology. Topology, under appropriate conditions, generates the metric geometry of spacetime.

Status

This chain is a hypothesis. The transitions $R \rightarrow D$ and $D \rightarrow \text{Adjacency}$ remain open problems from Volume I. The transition $\text{Adjacency} \rightarrow \text{Topology} \rightarrow \text{Geometry}$ is mathematically formalized in causal set theory and spin network approaches. The full chain is not yet closed from first principles.

ARTICLES XVI – XXI

Geometry as Modeling Infrastructure

Summary

Articles XVI through XXI investigate why self-modeling systems (S in the invariant sequence) generate geometric representations rather than some other representational structure. The convergent finding across six articles is:

Geometry is the most efficient representational structure for maintaining predictive compression of relational information. Self-modeling systems generate spacetime-like representations not because spacetime is the environment they inhabit but because geometry minimizes the representational overhead required to track relational persistence.

The specific chain is:

M → Compression → Locality → Geometry

Models compress reality. Compression requires locality — not all information can be equally relevant to all predictions. Locality requires neighborhood structure. Neighborhood structure induces geometry. The conclusion is that spacetime is a modeling solution rather than a fundamental substance.

The Observer Localization Corollary

Self-models require a distinction between inside and outside, a boundary, and a persistence criterion. These requirements jointly induce a coordinate system. Spacetime may therefore emerge as the bookkeeping system required for self-localization, appearing generically in any system that reaches the S transition of the invariant sequence.

Temporal Order as Constraint Propagation

Time appears wherever constraints propagate. Removing domain-specific vocabulary — future, past, before, after — leaves a single invariant: constraint update sequence. The hypothesis is that temporal order is an emergent property of constraint propagation rather than an independent primitive. This is consistent with the causal structure approaches in quantum gravity and with the internal time problem in canonical quantum cosmology.

Volume II — Summary Result

The Spacetime Audit finds no evidence that geometry occupies the deepest layer of explanation. Under UICP compression, spacetime consistently appears as an emergent structure generated by relational systems attempting to maintain distinctions, construct models, and localize self-representations. Geometry is neither fundamental nor terminal. It is an intermediate representational layer:

REVISED SEQUENCE — VOLUME II

R → D → M → Geometry → S → H

Spacetime is therefore interpreted not as the beginning of explanation but as a consequence of a deeper invariant process. The compression frontier has not moved. The sequence $R \rightarrow D \rightarrow M \rightarrow S \rightarrow H$ remains. Geometry has been located within it rather than beneath it.

VOLUME III

The Symmetry Audit

Broken Symmetry, Relational Structure, and the Origin of Distinction

Volumes I and II identified the sequence $R \rightarrow D \rightarrow M \rightarrow \text{Geometry} \rightarrow S \rightarrow H$ and located each transition with varying degrees of formal closure. Volume III pushes the audit one layer deeper by asking whether the origin of distinction — the unresolved $? \rightarrow D$ boundary — can be partially addressed by the existing physics of symmetry breaking.

ARTICLE XXIII

The Symmetry Compression Problem

Research Question

What survives UICP compression before distinctions exist?

Compression Result

Applying the protocol to the foundational structures of modern physics — gauge theories, general relativity, quantum field theory — yields a consistent residue: symmetry structures survive compression after objects, fields, and geometry have been removed.

Particles $\rightarrow$ **Fields** $\rightarrow$ **Geometry** $\rightarrow$ **Symmetry**

Gauge symmetries, Lorentz symmetry, diffeomorphism invariance, and internal gauge groups all survive stages of compression that eliminate their specific material content. The hypothesis following from this finding is that symmetry is deeper than object ontology.

ARTICLE XXIV

Distinction as Broken Symmetry

Observation

Before symmetry breaking, two states a and b are equivalent under the symmetry group: $a \equiv b$. After symmetry breaking, the equivalence is destroyed: $a \neq b$. A distinction has been generated.

Symmetry $\rightarrow$ **Broken Symmetry** $\rightarrow$ **Distinction**

Interpretation

Distinction emerges when equivalence classes become unstable under perturbation. This provides a partial mechanism for the $? \rightarrow D$ boundary identified in Article VII. Symmetry breaking does not derive distinction from something more primitive than distinction; it shows that distinction can emerge dynamically from a state of symmetric indistinguishability.

Prediction

Every distinguishable structure in the physical world should be traceable to a symmetry reduction event. This is consistent with the standard model, in which all particle species emerge from symmetry breaking of a higher-symmetry vacuum, and with spontaneous symmetry breaking in condensed matter, in which distinct phases emerge from a symmetric high-temperature state.

ARTICLES XXV – XXXIII

Gauge Structure, Information, and the Symmetry Horizon

Summary of Findings

Articles XXV through XXXIII audit the relationship between symmetry, relational structure, information, and the limit of the symmetry program. The principal findings are:

- Gauge fields, under UICP compression, appear less like forces and more like consistency-maintenance operators — correction terms ensuring that local transformations do not violate global relational coherence. The covariant derivative $D_\mu = \partial_\mu + igA_\mu$ is the formal implementation of this consistency requirement.
- Information requires distinction. Mutual information $I(A:B) = S(A) + S(B) - S(AB)$ is zero when A and B are indistinguishable. Information is therefore downstream of distinction in the invariant sequence: Symmetry $\rightarrow$ Distinction $\rightarrow$ Information.
- Multiple frontier approaches in quantum gravity (Ryu-Takayanagi, tensor network models, holographic entanglement entropy) derive geometric structure from entanglement structure. This is consistent with the UICP finding that relational structure precedes geometry: entanglement is a form of relational structure, and geometry emerges from it.
- Renormalization is a physical implementation of invariant-preserving compression. The renormalization group flow replaces microscopic detail with macroscopic invariants, which is the UICP compression operation applied to field-theoretic configuration spaces.

The Symmetry Horizon

The symmetry program partially explains the origin of distinction but does not eliminate the boundary. The problem is circular: defining symmetry requires specifying what may be transformed, which requires distinguishable states. The attempt to derive distinction from symmetry presupposes distinguishability at the level of the symmetry group itself.

? $\rightarrow$ **Symmetry**

This boundary — what generates symmetry structure without presupposing it — is the deepest unresolved question the UICP has reached. It is not answered here.

Volume III — Summary Result

The Symmetry Audit identifies symmetry as the strongest known precursor to the relational structure R that opens the invariant sequence. The revised hierarchy is:

REVISED SEQUENCE — VOLUME III									
?	→	Symmetry	→	R	→	D	→	M	→ Geometry → S → H

Current theoretical physics operates primarily within the region Symmetry → Geometry. The UICP audit extends the compression toward the unresolved transition ? → Symmetry, which now represents the deepest identified compression frontier of this research program.

Cross-Volume Synthesis

The Constraint Core: A Retroactive Unification

Reading across the three volumes reveals a pattern that was not visible within any single volume. Every framework audited — TNDS, functional invariance, adaptive multiplicity, Wavefront, Project Gamma, quantum error correction, renormalization, gauge theory, general relativity — reconstructs the same hidden object.

That object is not matter. Not energy. Not information. Not relation in the abstract. It is the minimal constraint set required to preserve system identity under perturbation. Designate it:

$$K = \text{Constraint Core}$$

Formally:

$$K = \text{minimal constraint set required to preserve system identity}$$

Rewriting the Corpus

Every major framework in this research program can be restated in terms of K:

Framework	Restatement in Terms of K
TNDS (survival)	A system survives when it reconstructs K faster than K decays
Functional invariance (M*)	M* counts alternative pathways for reconstructing K after perturbation
Wavefront (representation)	A useful representation is one that preserves K across translation
Project Gamma (substrate)	A substrate persists because downstream systems continuously reconstruct K through it
Soul Audit (selfhood)	Selfhood is recursive reconstruction — the system includes K in its own reconstruction target
Quantum error correction	Code distance d measures the number of errors correctable before K (the logical state) is destroyed
Renormalization	RG flow preserves K (macroscopic invariants) while discarding non-K microscopic detail
Gauge theory	Gauge fields maintain K (relational coherence) under local coordinate transformation
General relativity	Diffeomorphism invariance preserves K (geometric invariants) under coordinate change

The Unifying Question

The question the entire corpus has been asking, without naming it explicitly, is:

| *What must continue to be true for this system to remain itself?*

That question has a precise formal answer: K. The constraint core. And the answer is the same across every domain the compiler has audited.

Relation to the Invariant Sequence

K is not a new layer beneath the invariant sequence. It is the object being reconstructed at every arrow in the sequence. Relation generates distinction because relational structure is part of K. Distinction enables models because models are how systems track K. Self-models are systems that include K in their own reconstruction target. Horizons arise when K includes a representation of itself that cannot be fully self-described.

The invariant sequence $R \rightarrow D \rightarrow M \rightarrow \text{Geometry} \rightarrow S \rightarrow H$ is therefore best understood as a trajectory of increasing K-complexity: each transition corresponds to a system developing a more sophisticated mechanism for reconstructing K.

CENTRAL FINDING

K = the hidden common object the UICP has been locating across every domain

Relation to Prior Art

K is structurally closest to three existing constructs, none of which fully captures it:

- Aubin's viability theory formalizes the set of states a system must remain within to continue operating — this is K at the state-space level but does not address the reconstruction dynamics.
- Rosen's (M,R)-systems define life as a system that models its own repair — this captures the recursive reconstruction aspect of K but is scoped to biological systems.
- Ashby's requisite variety theorem states that a controller must have at least as much variety as the disturbances it faces — this captures the perturbation-resistance aspect of K but is formulated in cybernetic rather than constraint-theoretic language.

K synthesizes these three lineages into a single construct and generalizes across domains that none of them individually addresses. That generalization is the contribution of this synthesis.

Open Problems

The following problems are formally open at the close of Volume III. They are listed in order of depth, from most accessible to deepest.

1. Formalize the transition $D \rightarrow M$. Under what conditions does a collection of distinctions organize into a predictive model? The transition exists empirically — organisms and neural systems make it — but the mathematical conditions for when distinctions acquire predictive structure are not fully specified.
 2. Derive the transition $R \rightarrow D$ from first principles. Relational structure does not automatically generate distinction. Additional structure — instability, perturbation, asymmetry — is required. The formal conditions remain unspecified.
 3. Determine whether K is derivable from R , or whether K is the more fundamental object and R is downstream of it. The compression audit suggests R is deep. The cross-volume synthesis suggests K is deeper. Whether $K \rightarrow R$ or R is constitutive of K requires formal investigation.
 4. Formalize the transition $? \rightarrow \text{Symmetry}$. What structure generates symmetry without presupposing distinguishable transformations? This is the deepest boundary reached by the compiler. It may be that K itself is the answer — that the requirement that some distinctions persist is the pre-symmetry condition from which symmetry structures emerge. If so, the chain becomes self-grounding rather than regressive.
 5. Determine whether a deeper invariant exists beneath $R \rightarrow D \rightarrow M \rightarrow \text{Geometry} \rightarrow S \rightarrow H$. The compiler has reached $? \rightarrow \text{Symmetry}$ as the current frontier. Whether anything precedes the question mark, or whether the question mark is the terminal boundary of the method, is unknown.
 6. Derive the power-law relationship between M^* and reconstruction time from first principles within a specific domain. The qualitative relationship (larger kernel $\rightarrow$ faster reconstruction) is supported by multiple domain instances. A quantitative derivation in a single well-specified domain would constitute a publishable result independent of the broader program.
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Program Summary

Three volumes of recursive compression audit have produced the following results, listed by confidence:

Finding	Status
The UICP sequence $R \rightarrow D \rightarrow M \rightarrow \text{Geometry} \rightarrow S \rightarrow H$ is the strongest invariant currently extractable	Empirical finding; transitions $S \rightarrow H$ formalized; others open
Geometry is an intermediate emergent layer, not a primitive	Consistent with multiple independent physics programs
Symmetry is the deepest known precursor to relational structure R	Consistent with standard model and condensed matter physics
K (Constraint Core) is the hidden common object reconstructed across all audited domains	Synthesis finding; requires formal derivation
$? \rightarrow \text{Symmetry}$ is the current compression frontier	Unresolved; may require a framework not yet available

The search is not for a Theory of Everything. The search is for the deepest structure currently reachable by a compression method that takes no domain-specific vocabulary as given. That structure, at the close of this volume, is K — the constraint core — embedded within a sequence whose pre-history remains the outstanding open problem of the program.